

Billow

A minimum-energy structural model for soft kites
 Timoshenko beams, inflatable tubes, wrinkling membranes,
 cables and pulleys — one potential energy, one NLP

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Abstract

Billow is a standalone structural library for soft-wing airborne wind energy systems. It assembles cables, frictionless pulleys, geometrically exact Timoshenko beams, inflatable tubes and wrinkling membrane fabric into a single total potential energy, and finds static equilibrium by minimising it with IPOPT. Stationarity of the energy *is* the equilibrium equation, so a converged solve satisfies the nodal force balance to solver tolerance rather than to whatever residual a relaxation scheme happens to leave behind.

This document specifies the formulation, the element library, the validation against benchmarks external to the code, a set of demonstration cases, and a measured comparison against `kite_fem`, the existing FEM path. Headline results: exact-to- 10^{-4} agreement with the Euler elastica, second-order convergence to the exact roll-up circle, agreement with published values for the Bathe–Bolourchi 45° bend, reproduction of the ASKITE inflatable-tube fits to 0.06% in bending, and a full-kite structural solve in 5.9s cold and 0.58s warm at 1893 degrees of freedom.

Billow currently lives as `awetrim.structural` and imports only NumPy and CasADi. It is intended to become its own repository.

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1 Scope and design intent

1.1 What Billow is

A static structural solver for the kinds of structures soft kites are made of: thin lines that cannot push, ropes that run over sheaves, inflated tubes that bend until they collapse, and fabric that carries tension but wrinkles instead of carrying compression. Given a geometry, material properties and a set of nodal loads, it returns the deformed shape and the internal force distribution.

1.2 What Billow is not

It contains **no aerodynamics**. There are no polars, no airfoils, no lift model. Loads arrive as nodal forces from outside. Everything in this document that involves pressure uses a prescribed uniform differential pressure, not an aerodynamic solution. The same is true of `kite_fem`, which is also purely structural.

It is also not a dynamics code: there are no inertial terms and no time integration. Equilibrium is quasi-static.

1.3 Isolation

Billow depends only on NumPy and CasADi. It does not import the particle-system solver (PSS), the vortex-step method (VSM), the aerostructural coupling loop or any kite YAML schema, and nothing in those modules imports it. This is deliberate and is the first invariant to preserve: it allows the element physics to be validated against closed-form solutions before any of it touches a coupled run. A coupling adapter, when written, belongs on the aerostructural side, not inside Billow.

1.4 Why a new model

The existing minimum-energy solver in the AWETrim tree (`aerostructural/pss/structural_nlp.py`) already solves the current bridle well. It does not extend, for two independent reasons.

Physics. Its element vocabulary is axial springs. Battens and leading-edge tubes need bending and torsion; the canopy needs a membrane that cannot carry compression.

Scale. It builds the objective with a Python loop over elements in CasADi `MX`, so the expression graph grows with the element count. At tens of springs this is free. At tens of thousands of fabric triangles, graph construction — and especially `gradient` and `hessian` over it — becomes the bottleneck long before the numerical solver does.

Billow keeps the formulation (still an NLP, still IPOPT, still minimum potential energy) and replaces the assembly.

2 Formulation

2.1 Total potential energy

The unknown configuration X minimises

$$\Pi(X) = \sum_{\text{sets}} U_{\text{set}}(X) - \mathbf{f}_{\text{ext}} \cdot \mathbf{x} - \mathbf{m}_{\text{ext}} \cdot \boldsymbol{\theta}(\boldsymbol{\psi}) + \frac{1}{2} k_a \|\mathbf{x} - \mathbf{x}_{\text{seed}}\|^2, \quad (1)$$

subject only to bounds on pinned degrees of freedom. The last term is a weak anchor discussed in §2.8.

Because $\partial\Pi/\partial X = 0$ is the equilibrium equation, a converged solve satisfies $\mathbf{f}_{\text{int}} + \mathbf{f}_{\text{ext}} = 0$ at every free node to solver tolerance. There is no separate residual criterion to tune.

2.2 Degrees of freedom

$$X = \left[\underbrace{\mathbf{x}_0 \dots \mathbf{x}_{n-1}}_{3 \text{ per node}} \mid \underbrace{\boldsymbol{\psi}_0 \dots \boldsymbol{\psi}_{r-1}}_{3 \text{ per rotational node}} \right] \quad (2)$$

Three translations for *every* node, and three incremental rotations only for the r nodes that some beam element needs a material frame for. Cable, pulley and membrane nodes stay at three degrees of freedom. A canopy of 10 000 fabric nodes therefore costs 30 000 unknowns, not 60 000, regardless of what beams are attached elsewhere. For contrast, `kite_fem` allocates six degrees of freedom per node everywhere and masks the unused ones.

2.3 Finite rotations: the Cayley parameterisation

Rotations use the **Rodrigues (Cayley) vector** $\boldsymbol{\psi} = 2 \tan(\phi/2) \mathbf{n}$ rather than the exponential map's rotation vector $\phi \mathbf{n}$. With $S = \text{skew}(\boldsymbol{\psi})$,

$$R(\boldsymbol{\psi}) = I + \frac{4}{4 + \boldsymbol{\psi} \cdot \boldsymbol{\psi}} (S + \tfrac{1}{2} S^2), \quad (3)$$

$$\boldsymbol{\psi}(R) = \frac{2 \text{ axial}(R - R^\top)}{1 + \text{tr } R}, \quad (4)$$

$$\boldsymbol{\psi}_{1/2} = \frac{\boldsymbol{\psi}}{1 + \sqrt{1 + \tfrac{1}{4} \boldsymbol{\psi} \cdot \boldsymbol{\psi}}}, \quad R(\boldsymbol{\psi}_{1/2})^2 = R(\boldsymbol{\psi}). \quad (5)$$

The motivation is entirely numerical. Every map above is a *rational* function of its argument. There is no $\sin \phi / \phi$ removable singularity, so no `if_else` branch is needed at $\phi = 0$, so the Hessian that IPOPT differentiates is continuous everywhere in $|\phi| < \pi$. The half-angle identity (5) is worth noting: the naive form $\tan(\phi/4) = (\sqrt{1+t^2} - 1)/t$ has a removable $0/0$ at $t = 0$; rationalising it removes the singularity exactly.

2.4 Incremental rotations and frame updates

$\boldsymbol{\psi}_j$ is measured against a stored reference frame, so the configuration is $R_j = R(\boldsymbol{\psi}_j) R_{\text{ref},j}$ and every solve starts from $\boldsymbol{\psi} = 0$. After each solve the increment is absorbed, $R_{\text{ref}} \leftarrow R(\boldsymbol{\psi}^*) R_{\text{ref}}$, and re-projected onto $SO(3)$ so repeated absorption cannot drift.

The Cayley chart is singular at $\phi = \pi$. Incremental rotations mean a node can turn arbitrarily far *across a load ramp* while each individual solve stays well inside the chart. This is why applied moments require load stepping (§2.5); nodal forces do not.

2.5 Applied moments are not conjugate to $\boldsymbol{\psi}$

A moment does work against the *exponential-map* rotation vector, not against the Rodrigues vector the degrees of freedom carry. The external work term therefore converts,

$$\boldsymbol{\theta}(\boldsymbol{\psi}) = \boldsymbol{\psi} \frac{2 \arctan(\|\boldsymbol{\psi}\|/2)}{\|\boldsymbol{\psi}\|}, \quad (6)$$

which tends to $\boldsymbol{\psi}$ as $\|\boldsymbol{\psi}\| \rightarrow 0$.

This is not cosmetic. Using $-\mathbf{m} \cdot \boldsymbol{\psi}$ directly is correct only to first order, and the roll-up benchmark (§4.2) exposed it as a *load-path dependence*: the same static problem gave $8.1 \times 10^{-3} L$ with four load steps and $2.1 \times 10^{-1} L$ with eight. Statics must be path independent. With the conversion in place the answer is identical to five digits across 4, 8 and 16 steps, and this is guarded by a regression test.

Two limitations remain, both fundamental rather than implementation artefacts:

- Applied moments need load stepping, because the chart is singular at $\phi = \pi$. Four steps suffice for a full 2π roll-up.
- A constant (“dead”) moment in three dimensions is **genuinely non-conservative** and therefore has no potential at all. Energy minimisation can represent only the fixed-axis case.

Nodal *forces* — which is what an aerodynamic coupling applies — carry none of these caveats and are exact in a single solve.

2.6 Assembly: one mapped kernel per element type

This is the single design decision that makes the model scale. Each element kernel is compiled once to a small CasADi `SX` function over *one* element and evaluated across the whole set with `Function.map` on a gathered index matrix:

$$Q = \text{reshape}(X[\text{idx}], n_{\text{dof/el}}, n_{\text{el}}), \quad U_{\text{set}} = \sum_e \mathcal{U}(Q_{:,e}, F_{:,e}, P_{:,e}).$$

The objective graph therefore holds one node per element *type*, not per element. Graph construction, gradient and Hessian generation cost $O(\text{number of element types})$, not $O(\text{number of elements})$.

The Hessian retains the sparsity of the element connectivity graph, so IPOPT’s sparse linear algebra sees a mesh-stencil matrix and scales accordingly. The measured consequence is Table 1.

Table 1: Flat clamped canopy, uniform out-of-plane load. Iteration count is essentially mesh independent; cost sits in the sparse linear algebra. Source: `run_demo_cases.py --case scaling`.

Triangles	DOF	Build (s)	Solve (s)	IPOPT iterations
128	243	0.07	0.05	6
1800	2883	0.67	1.25	9
5832	9075	3.0	6.1	9

2.7 Everything numeric is a parameter

Reference frames, every element parameter column (rest lengths, stiffnesses, section properties), the external loads and the anchor are NLP *parameters*, not baked constants. One solver build therefore serves an entire sweep: tape actuation, a stiffness ramp or a load sweep change parameter values, never topology.

2.8 The anchor term

The final term of (1) is a weak spring to the seed positions, present only to pin a node that no taut element reaches and whose equilibrium position would otherwise be indeterminate. It biases the reported force balance by at most $k_a \|\mathbf{x} - \mathbf{x}_{\text{seed}}\|$; the default $k_a = 10^{-6} \text{ N/m}$ puts that far below any coupling tolerance.

2.9 Convergence criterion

`converged` is the *physical* verdict: the solve is accepted when the free nodes are in force balance to

$$\max(\varepsilon_{\text{abs}}, \varepsilon_{\text{rel}} \max_i \|\mathbf{f}_{\text{ext},i}\|), \quad \varepsilon_{\text{abs}} = 10^{-6} \text{ N}, \quad \varepsilon_{\text{rel}} = 10^{-6},$$

whether or not IPOPT liked its own termination test. The relative term matters: an absolute newton means very different things on a 1 N bridle line and a 600 N benchmark beam. IPOPT’s own verdict is preserved alongside as `ipopt_success` and `status` — on stiff, nearly quadratic structures it routinely reports `Search_Direction_Becomes_Too_Small` at an answer already exact to machine precision.

3 Element library

Every kernel returns the strain energy of one element. Kernels are pure: no loops over elements, no global indexing, no closing over per-element data. Everything element-specific arrives through the parameter row.

3.1 Cable

A linear axial spring with stretch $s = \ell - \ell_0$:

$$U = \frac{1}{2}k \langle s \rangle^2, \quad \langle s \rangle = \max(0, s) \text{ when tension-only.} \quad (7)$$

The tension-only cut is C^1 but not C^2 : the stiffness jumps across the taut/slack boundary. An optional `slack_smoothing` width rounds the corner to recover a continuous Hessian at the cost of a small spurious compressive stiffness.

3.2 Pulley

A rope running over a frictionless sheave at the middle node, with a single shared stretch across both arms:

$$U = \frac{1}{2}k \langle \|\mathbf{x}_2 - \mathbf{x}_1\| + \|\mathbf{x}_3 - \mathbf{x}_2\| - \ell_0 \rangle^2. \quad (8)$$

One shared stretch is exactly the statement that rope tension is equal either side of the sheave, and it lets the pulley node find its own position along the rope.

Convention warning. `rest.length` here is the **whole rope**, both arms. This is not universal: the PSS geometry reader splits ℓ_0 across the two arms, while `kite.fem` stores the total on each arm. Any geometry adapter must state which convention it is reading.

3.3 Geometrically exact Timoshenko beam

Two-node Simo–Reissner beam with one-point (mid-element) integration. Each node carries a position and an orthonormal material frame $R = [\mathbf{d}_1 \mathbf{d}_2 \mathbf{d}_3]$: twelve degrees of freedom per element.

With ψ the relative rotation between the two node frames from (4), and R_m the exact half-way frame from (5):

$$\mathbf{\Gamma} = R_m^\top (\mathbf{x}_2 - \mathbf{x}_1) / L_0 \quad \text{axial stretch and two transverse shears} \quad (9)$$

$$\mathbf{\Omega} = R_m^\top \psi / L_0 \quad \text{torsion and two bending curvatures,} \quad (10)$$

$$U = \frac{L_0}{2} \left[\Delta \mathbf{\Gamma}^\top \text{diag}(EA, GA_2, GA_3) \Delta \mathbf{\Gamma} + \Delta \mathbf{\Omega}^\top \text{diag}(GJ, EI_2, EI_3) \Delta \mathbf{\Omega} \right], \quad (11)$$

with $\Delta(\cdot) = (\cdot) - (\cdot)_0$ measured against the reference configuration, so an initially curved or pre-twisted member is stress-free as built.

Three properties make this element usable inside an NLP objective:

Objective. Under a superposed rigid rotation Q , both ψ and $\mathbf{x}_2 - \mathbf{x}_1$ map to $Q(\cdot)$ while $R_m \rightarrow QR_m$, so $\mathbf{\Gamma}$ and $\mathbf{\Omega}$ are invariant and rigid-body motion costs exactly zero energy. A slack member that swings freely is not spuriously stiffened.

Shear-locking free. The single midpoint sample is the standard reduced integration cure. Kite battens and leading-edge tubes are slender, so this is not optional; §4.5 verifies it at slenderness 800.

Branch-free. Using the Cayley vector as the curvature measure makes the element strain $\mathbf{\Omega}_{\text{exact}} + O(\phi^3)$ per element — the same order as the one-point quadrature error, and vanishing under mesh refinement. This is a deliberate trade: the exponential map’s $\sin \phi / \phi$ would need a branch at $\phi = 0$.

3.4 Inflatable tube

An inflated leading-edge tube is not linear-elastic. Its bending stiffness falls away as the compressed side wrinkles, until the section carries a limiting moment and then collapses. `kite_fem` captures this with empirical fits (coefficients $C_1 \dots C_{19}$, from the ASKITE work) applied as *secant* stiffnesses: EI and GJ are recomputed each iteration from the current deflection and twist.

That is consistent inside a Newton residual, but it is **not a potential**. Substituting a state-dependent $EI(\kappa)$ into $\frac{1}{2}EI\kappa^2$ silently drops the $dEI/d\kappa$ terms, so the gradient would no longer be the internal moment and a minimum-energy solve would converge to the wrong equilibrium. The fits must be *integrated*, not substituted.

From tip-load fit to moment–curvature law. The bending fit is published as tip load against normalised tip deflection $v = \delta/L$ of a **one-metre** cantilever: $P(v) = D(1 - e^{-(N/D)v})$. For a linear cantilever $\delta = PL^3/3EI$ and $\kappa_{\text{root}} = PL/EI$, so $\kappa = 3v/L$, and the root moment is $M = PL$. At the calibration length $L = 1$ m this converts the fit into an intrinsic constitutive law:

$$M(\kappa) = M_{\text{max}} \left(1 - e^{-EI_0\kappa/M_{\text{max}}}\right), \quad EI_0 = N/3, \quad M_{\text{max}} = D, \quad (12)$$

$$W_b(\kappa) = M_{\text{max}} \left[|\kappa| - k_0 \left(1 - e^{-|\kappa|/k_0}\right)\right], \quad k_0 = M_{\text{max}}/EI_0. \quad (13)$$

W_b is quadratic ($\frac{1}{2}EI_0\kappa^2$) near zero, so it is smooth there despite the $|\kappa|$, and saturates at M_{max} .

Recasting this way also removes a length inconsistency: `kite_fem` infers $EI = P/(3v)$, which equals the true EI only for a one-metre element, because v is already normalised by element length. A moment–curvature law is length-independent by construction.

Torsion is already intrinsic (twist per unit length) and integrates directly:

$$T(\omega) = c_1 \arctan(c_2\omega), \quad W_t(\omega) = c_1 \left[\omega \arctan(c_2\omega) - \frac{\ln(1 + c_2^2\omega^2)}{2c_2} \right], \quad (14)$$

with $GJ_0 = c_1c_2$ and a limiting torque $c_1\pi/2$.

Biaxial bending uses the resultant curvature $\sqrt{\Omega_2^2 + \Omega_3^2}$, the correct isotropic extension for an axisymmetric tube. Axial and shear stiffness are *not* covered by the fits and stay linear; supply them from the tube geometry.

Table 2: Derived tube properties from the ASKITE fits. Higher pressure stiffens and strengthens; the thinner tube collapses *before* its moment saturates ($\kappa_{\text{collapse}} < k_0$).

d (m)	p (bar)	EI_0 (N m ²)	M_{max} (N m)	k_0 (m ^{−1})	κ_{collapse} (m ^{−1})	GJ_0 (N m ²)	T_{max} (N m)
0.16	0.3	288.2	58.10	0.2016	0.2751	91.3	124.9
0.16	0.5	362.0	98.78	0.2728	0.2973	86.5	174.7
0.12	0.3	126.8	34.16	0.2694	0.2217	33.2	117.1

Collapse is reported, not enforced. A genuinely dropping post-collapse moment would make the energy *decrease* with curvature — an unbounded mechanism that minimisation would simply run away from. The law therefore stops at saturation, and `inflatable_beam_state` returns a utilisation ratio $\kappa/\kappa_{\text{collapse}}$ and a `collapsed` flag, so leaving the calibrated range is visible rather than silent.

3.5 Wrinkling membrane

Constant-strain triangle with a Saint-Venant–Kirchhoff plane-stress law, in the finite-strain (Green–Lagrange) measure so a canopy panel can rotate and billow arbitrarily without spurious stiffening. Translation-only: a membrane carries no bending.

The reference triangle is flattened once into its own in-plane basis and stored as the inverse edge matrix D_0^{-1} , so the kernel only ever sees a 2×2 map:

$$F = [\mathbf{a} \ \mathbf{b}] D_0^{-1}, \quad C = F^\top F, \quad E = \frac{1}{2}(C - I), \quad (15)$$

$$W_{\text{taut}} = \frac{E_{\text{mod}}}{2(1 - \nu^2)} \left[(\text{tr } E)^2 - 2(1 - \nu) (E_{11}E_{22} - E_{12}^2) \right]. \quad (16)$$

Wrinkling. A membrane cannot carry compression. Left alone, the SVK law lets the canopy mesh buckle into whatever compressive mode the discretisation offers, and the result depends on mesh density rather than on physics. The relaxed (tension-field) energy in principal Green strains $e_1 \geq e_2$ is the classical Pipkin form:

$$W_{\text{relaxed}} = \begin{cases} W_{\text{taut}}, & e_2 + \nu e_1 \geq 0 \quad (\text{taut}), \\ \frac{1}{2} E_{\text{mod}} e_1^2, & e_1 > 0 \text{ otherwise} \quad (\text{wrinkled}), \\ 0, & e_1 \leq 0 \quad (\text{slack}). \end{cases} \quad (17)$$

This is exactly what a minimum-energy formulation wants: the relaxed functional is the quasiconvex envelope, so the minimiser lands on the wrinkled state directly instead of chasing the near-singular tangent a residual-form Newton solve has to fight through. **Solving fabric by energy minimisation is not a workaround — it is the better-posed formulation.**

Slack fabric needs a residual stiffness. A fully slack region stores *exactly* zero energy, so its Hessian block is exactly zero and the Newton step is undefined; IPOPT reports `Error_In_Step_Computation` on a panel clamped into a frame smaller than itself. Blending a small fraction of the unrelaxed law back in,

$$W = (1 - r) W_{\text{relaxed}} + r W_{\text{taut}}, \quad r = 10^{-4}, \quad (18)$$

leaves the taut region untouched (there the two coincide) while giving slack fabric a definite tangent. It is also not a fiction: real fabric has some compressive and bending resistance. Do not set $r = 0$ on a canopy that can go slack.

4 Validation

All references in this section come from *outside* the code: an exact boundary-value solution, an exact circle, published tabulated values, or the independent fits themselves. Nothing here can drift with the implementation.

Where `kite_fem` appears it is driven with matched linear section properties and its inflatable constitutive update disabled, so the two share one material model. Section properties are deliberately slender: the classical references are inextensible and shear-rigid, so axial and shear flexibility are kept below $\sim 0.1\%$ and cannot be mistaken for discretisation error.

4.1 Euler elastica

A tip-loaded cantilever at load parameter $\alpha = PL^2/EI$ up to 10. The reference is the exact elastica, obtained by solving $\theta'' = -\alpha \cos \theta$, $\theta(0) = 0$, $\theta'(1) = 0$ as a boundary-value problem to 10^{-10} with continuation in α — no beam element is involved.

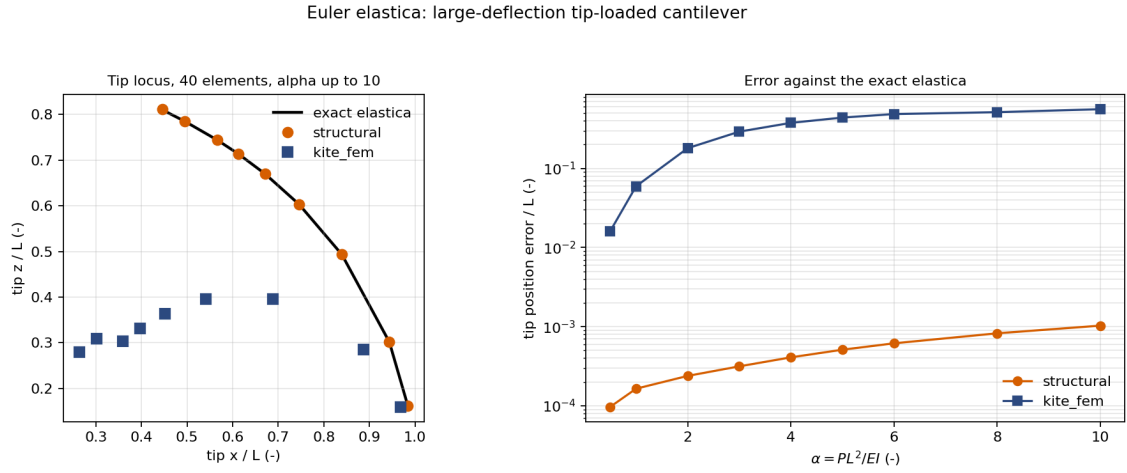


Figure 1: Tip locus and error against the exact elastica, 40 elements. Billow tracks the exact curve across the whole range; `kite_fem` departs progressively and its tip deflection saturates beyond $\alpha \approx 2$.

Table 3: Tip position error normalised by length, 40 elements.

α	0.5	2	6	10
Billow	9.0×10^{-5}	2.4×10^{-4}	6.5×10^{-4}	1.1×10^{-3}
kite_fem	1.6×10^{-2}	1.8×10^{-1}	4.2×10^{-1}	4.9×10^{-1}

Mesh refinement separates discretisation from formulation. At $\alpha = 3$, refining from 10 to 160 elements:

Elements	10	20	40	80	160
Billow	3.8×10^{-3}	7.0×10^{-4}	3.1×10^{-4}	4.6×10^{-4}	5.0×10^{-4}
kite_fem	2.908×10^{-1}	2.914×10^{-1}	2.916×10^{-1}	2.917×10^{-1}	2.917×10^{-1}

The `kite_fem` error is **mesh independent**, which places it in the formulation rather than the discretisation. Its beam matches linear theory to 0.02% below about one degree of tip rotation, so the property matching used here is correct; the error is purely rotation-driven. Envelope: below 1% up to roughly 10° , about 15% at $\alpha = 1$, saturating beyond $\alpha = 2$. It reports `converged=True` at those answers.

4.2 Roll-up under an end moment

A cantilever under $M = 2\pi EI/L$ must roll into an exact closed circle, so the reference requires no lookup at all.

Roll-up under an end moment: the exact answer is a closed circle

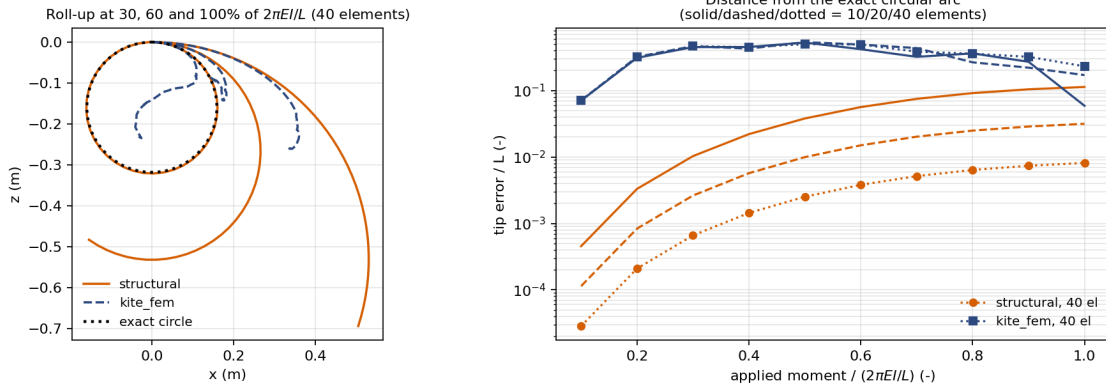


Figure 2: Roll-up at 30, 60 and 100% of $2\pi EI/L$, and the distance from the exact circular arc. Billow converges at second order; `kite_fem` moves away from the exact answer under refinement.

Table 4: Tip distance from the exact arc, normalised by length.

Elements	10	20	40
Billow (4 load steps)	1.13×10^{-1}	3.16×10^{-2}	8.14×10^{-3}
<code>kite_fem</code> (single shot)	5.90×10^{-2}	1.71×10^{-1}	2.32×10^{-1}

Billow's error ratios are 3.6 and 3.9, i.e. second order. Extending to 80 elements gives 2.05×10^{-3} , confirming the trend. This benchmark is what exposed the moment-conjugacy defect of §2.5.

4.3 Bathe–Bolourchi 45° bend

The standard three-dimensional finite-rotation benchmark: a 45° circular arc of radius 100, square section, out-of-plane tip load $F = 600$. Published tip *displacements* differ by about 1% between authors, so the reference is a band.

Bathe & Bolourchi 45-degree curved cantilever, out-of-plane load $F = 600$ (tip displacement)

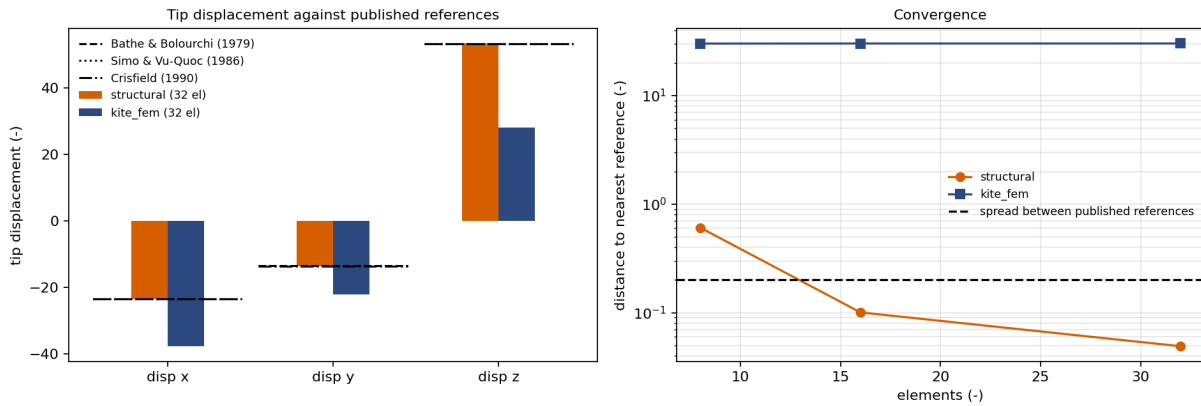


Figure 3: Tip displacement against published references, and convergence.

Table 5: Tip displacement, 32 elements. The arc here starts at the origin with its tangent along $+x$ and sweeps towards $+y$, which swaps the two in-plane components relative to the usual tabulation.

	u_x	u_y	u_z	$\ \mathbf{u}\ $
Simo & Vu-Quoc (1986)	-23.48	-13.50	53.37	59.85
Bathe & Bolourchi (1979)	-23.5	-13.4	53.4	59.85
Crisfield (1990)	-23.5	-13.6	53.4	59.88
Billow	-23.52	-13.60	53.44	59.95
<code>kite_fem</code>	-37.74	-22.14	28.17	52.04

Billow sits within the spread between published references (0.17% on the magnitude). `kite_fem` is 47% low out of plane.

4.4 Inflatable tube law

The port of §3.4 is verified with **pure end moments**, where the exact solution is a constant-curvature arc at whatever curvature the fit prescribes. That check is exact at any deflection — unlike the tip-load form the fit was published in, whose own inversion assumes linear cantilever theory and therefore cannot serve as a verification instrument.

Inflatable tube: ASKITE fits recast as a strain energy, verified by pure end-moment solves

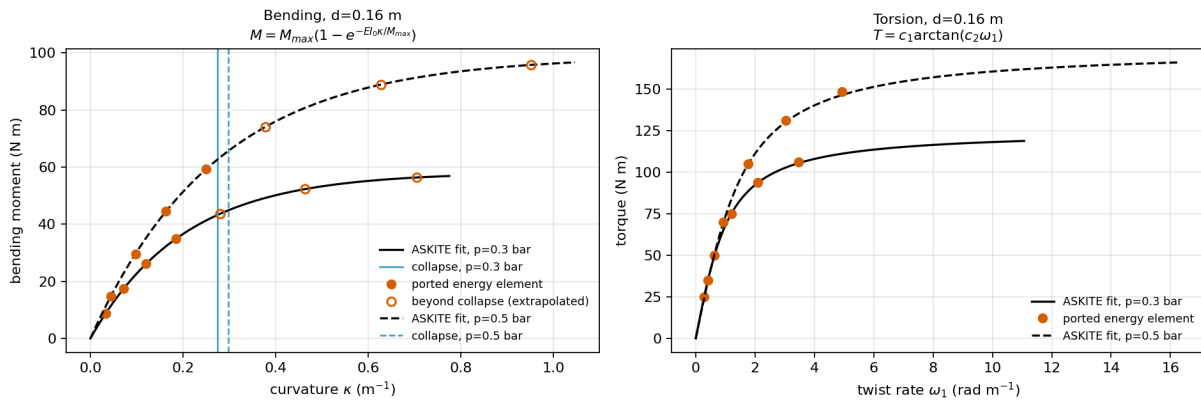


Figure 4: ASKITE fits (lines) with Billow end-moment solves on top (markers). Hollow markers lie beyond the collapse curvature, where the fit is extrapolation: the solve is fine, the calibration is not. This figure completes `kite_fem/examples/FEM_beam_verification.py`, which plots the fitted curves but leaves the simulation loop as comments.

Worst errors: 0.031% (bending, 0.3 bar), 0.057% (bending, 0.5 bar), 0.749% and 1.491% (torsion).

4.5 Beam convergence and shear locking

Tip deflection of a slender cantilever must approach $PL^3/3EI + PL/\kappa GA$ at second order:

Elements	2	4	10	20	40	80
Error vs Timoshenko	-6.25%	-1.56%	-0.250%	-0.063%	-0.016%	-0.004%

The error falls by a factor of four per refinement. Locking is checked separately by holding the load and lengthening the beam through slenderness ratios of 50, 200 and 800: the tip deflection tracks beam theory to 0.2% throughout, where a locking element would stiffen towards zero.

4.6 Membrane

Checked against closed form rather than against another code:

- **Rigid-body motion** costs zero energy under arbitrary rotation and translation — what the finite-strain measure buys over a small-strain one.
- **Equibiaxial stretch** matches the analytic SVK plane-stress energy to 10^{-9} relative.
- **Wrinkled branch** is exactly uniaxial: stretching along x while squashing along y gives $\frac{1}{2}EtAc_1^2$, and squashing further changes nothing.
- **Slack panel** stores exactly zero, where the unrelaxed law would store large energy.
- **Flat-plate inflation** converges under mesh refinement from a configuration with zero out-of-plane stiffness — the singular start that a residual-form Newton method has nothing to invert at.

4.7 Test suite

86 tests, all passing, organised one file per concern:

<code>test_rotations.py</code>	Cayley round trips, half-angle identity
<code>test_cable.py</code>	Hooke parity with PSS, slack cut, pulley tension equalisation
<code>test_beam.py</code>	objectivity, $O(h^2)$ convergence, no locking, torsion vs GJ
<code>test_membrane.py</code>	analytic SVK, three wrinkling branches, inflation, scale
<code>test_inflatable.py</code>	energy integrates the fitted moment; end-moment solves
<code>test_benchmarks.py</code>	roll-up vs exact circle, load-step independence, bend45
<code>test_energy.py</code>	mapped assembly vs a naive per-element sum; DOF layout

The most important of these is in `test_energy.py`: the total energy computed through the gather-and-map machinery is compared against the same energy assembled one element at a time in plain Python, on a deliberately irregular model containing every element type with a beam running over a scattered, out-of-order subset of the node numbering. That machinery is easy to get subtly wrong — a column-major reshape, a rotational slot off by one — and wrong in a way that still converges to a plausible-looking shape.

5 Validation against a measured kite

The benchmarks above are exact solutions of idealised problems. This section is the only place where Billow meets a real structure with measured deformation: the TU Delft V3 suspended from a bridle bar in a hangar, at two inflation pressures and five loadings, with its deformed shape recorded by stereoscopic photogrammetry. Data and reference solution come from `awegroup/kite_fem` (MIT) and Roeleveld’s MSc thesis; the 3D marker clouds come from Haanen’s repository.¹ A full account is in `docs/hanging-validation/`; what follows is the part that bears on Billow itself.

5.1 The model, and what had to be fixed in the reference

Billow is rebuilt from the reference `.npz` rather than from YAML, so nodes, connectivity, rest lengths, stiffnesses and tube diameters transfer with no re-interpretation: 276 nodes, 46 bridle cables, 14 pulleys, 98 inflatable tube beams shared element for element, with only the canopy changed (721 tension-only springs become 392 wrinkling membrane triangles). The canopy is a

¹https://github.com/pimjhaanen/photogrammetry_thesis.
data/LEI-V3-KITE/hanging-validation/.

structured 29×8 grid recovered geometrically from a flat spring list and then *verified* — all 721 canopy springs must come out as grid edges.

Two defects in the distributed reference had to be dealt with. Its anchor node is flagged fixed but belongs to no element, so the kite is a free body under 126 N of unbalanced weight and the potential is unbounded below; Billow reports this faithfully by translating the structure hundreds of kilometres. Roeleveld’s Appendix B supplies the missing suspension — a 1650 mm bar across the two main bridle convergence points — and eight of the thirty-six bridle lines disagree with his Table B.1. Both are documented in `scripts/structural/hanging_kite.py`.

5.2 What the measurement says about the tube law

Only two of the ten cases actually load the tubes. Wherever an applied load dominates, the bridle carries it and the shape is fixed kinematically by bridle lengths, which both codes read from the same file: the two answers then agree to within 1–300 mm. Under gravity alone they differ by more than a metre. The photogrammetry confirms this independently — on the tip-load cases the *undeformed* geometry scores as well as any model.

On the two cases that do test the tubes, Billow’s wing stays too open. Scaling the whole moment–curvature curve by a single factor (`scale_tube_laws`, which moves $M_{\max}$, EI_0 and the torsion constant together so κ_0 is unchanged) the measurement implies $\lambda \approx 0.36$, and implies the *same* factor at both pressures. Against the 3D markers:

mean point-to-surface RMS (mm)	as built	$EI \times 1.0$	$EI \times 0.36$
two gravity cases	625	434	101
four centre-load	583	454	155
four tip-load	115	117	108
all ten	404	315	125

The factor was identified from two scalar spans and then reproduced the full three-dimensional shape, including the four centre-load cases it was not fitted on, bringing the tube-loaded cases onto the ~ 100 mm marker floor. That generalisation is why it looks like a property of the tube law rather than a number absorbing error.

It is not adopted, and why. Inverting $EI = Et\pi r^3$ gives an implied Et of 133–155 kN/m, near-constant across $d = 0.113 - -0.202$ m — so the fit’s radius dependence is sound and the implied fabric is plausible. The likelier explanation is a mismatch in how the fit is being *re-used*: Breukels calibrated it on rigid segments joined by torque springs rather than as a continuum law, his nomenclature distinguishes an effective internal pressure from the gauge value, and the softening in his cantilever data is structural (a spreading wrinkle zone) rather than pointwise. All three would produce a constant factor, and all three should be checked before Billow’s tube law is changed.

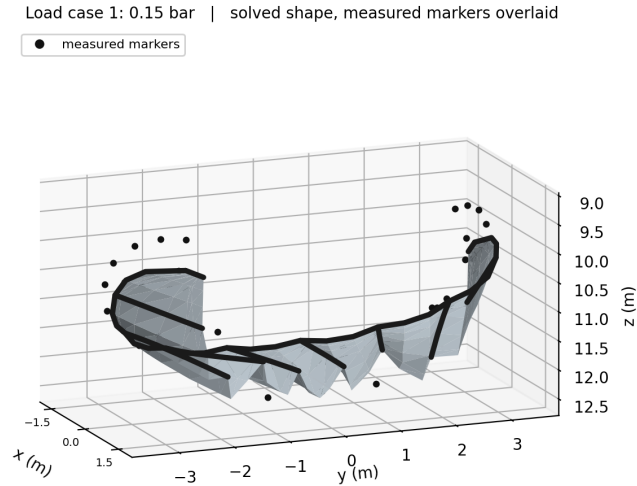


Figure 5: The V3 solved under self-weight, with Haanen’s photogrammetry markers overlaid. Shading is by facet orientation, so the billow between struts reads directly.

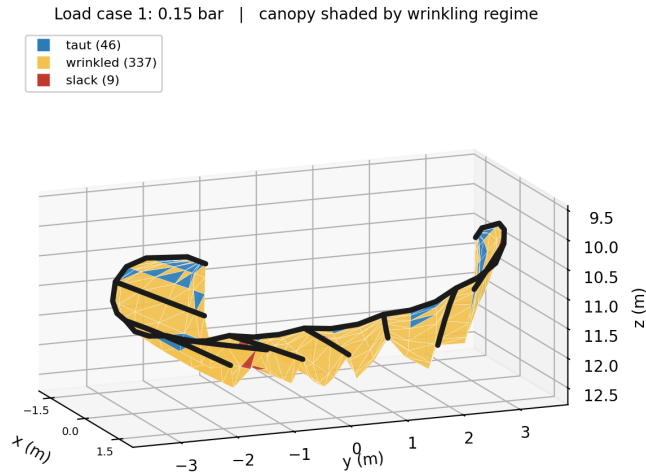


Figure 6: The same solution with each canopy triangle coloured by tension-field state. Almost the whole canopy is wrinkled under self-weight, taut only where the bridle pulls through. A tension-only spring net has no equivalent prediction.

5.3 Collapse, and what it means for coupled use

Section 3.4 notes that collapse is reported rather than enforced. This case settles how much that matters: the beams here run at a median κ/κ_0 of 0.097 and $M/M_{\max}$ of 0.093, i.e. at 95% of their initial stiffness, and none of the 98 is flagged collapsed. A law that genuinely follows collapse, $W = EI_0\kappa_c^2[1 - (1 + \kappa/\kappa_c)e^{-\kappa/\kappa_c}]$, converges without difficulty (56–77 iterations) and changes the answer by 20 mm.

A flying kite carries a distributed, bridle-reacted load and keeps its tubes on the same initial slope, so saturation is adequate for aero-structural use — not because collapse is handled but

because it is not reached. What governs a coupled run is EI_0 , which is exactly the quantity this dataset questions.

5.4 Cost, and a caveat on the comparison

Billow reaches force balance to better than 5×10^{-7} N on all ten cases in a mean of 14 s. `kite_fem`'s recorded times average 552 s, and three of its cases stop above its own 0.01 tolerance.

That ratio should be read as an order of magnitude and no more. The two sets of times are from different machines, the two codes are not solving to the same tolerance, and — more importantly — **kite_fem was never re-run for this study**: every figure attributed to it comes from its committed results. Its converged beams also sit at about $0.23\times$ its own constitutive law, with negative bending stiffness in several cases, so it is not a clean reference solution. The full list of caveats is in `docs/hanging_validation/`, and they should be read before any comparative claim here is repeated.

5.5 What this dataset cannot decide

The comparison was built to isolate the canopy formulation, the one element type that differs between the codes. It cannot do that: deleting the canopy entirely moves the span by 0.17 m, while a modest change in tube bending moves it by 2.5 m. It is a tube-bending test, and no canopy conclusion should be read off it.

The four centre-load cases are outside what any beam element can represent: local leading-edge indentation is a cross-section collapse, and a one-dimensional moment–curvature element has no degrees of freedom for it. Roeleveld reaches the same conclusion about his own model.

6 Demonstration cases

6.1 Wrinkling: fabric in a frame smaller than itself



Figure 7: A panel clamped into a frame 6% smaller than itself, then pushed out of plane. The relaxed law converges in 31 iterations to a smooth billow; the unrelaxed one takes 211 and buckles into mesh-scale crumple at less than two-thirds the deflection.

This is the clearest illustration of why the relaxed energy is not optional. The unrelaxed panel stores 40.6 J against the relaxed panel's 0.116 J, almost all of it fighting compression that fabric cannot carry.

6.2 Large-deflection cantilever

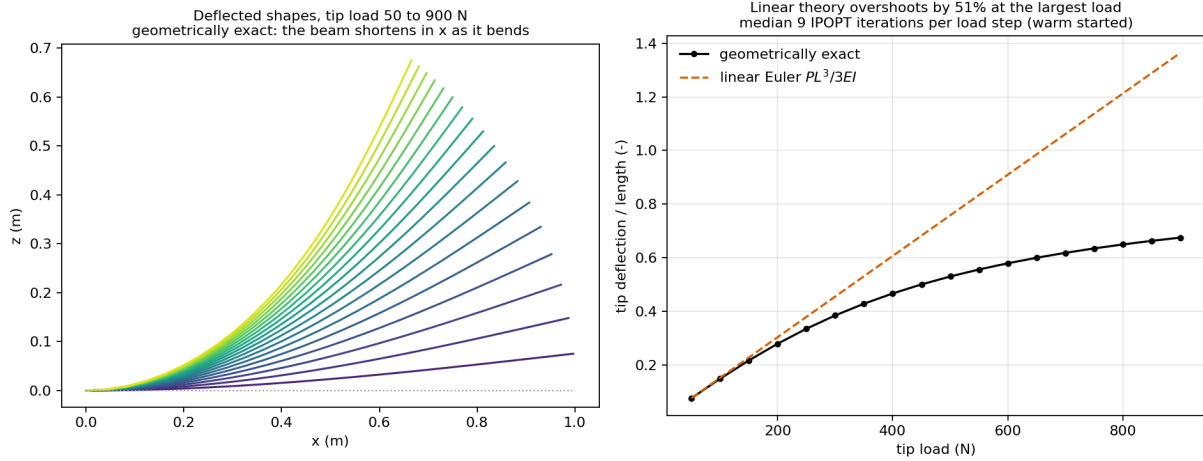


Figure 8: Deflected shapes under a tip load swept from 50 to 900 N, and tip deflection against linear theory. The beam visibly shortens in x as it bends — the geometric nonlinearity linear theory omits. At the largest load, linear theory overshoots by 51%.

Median 9 IPOPT iterations per load step with warm starting.

6.3 Batten-stiffened sail

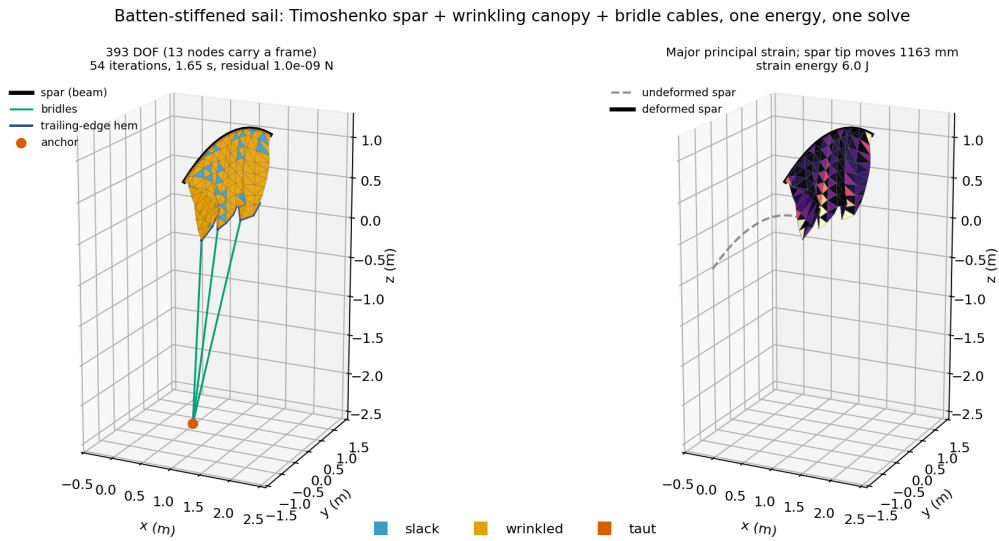


Figure 9: Beams, membranes, cables and a trailing-edge hem in one energy: a spar carrying bending and torsion, a wrinkling canopy hanging off it, three bridles to a fixed anchor. 393 DOF, 54 iterations, 1.4 s, residual 1.0×10^{-9} N.

Note that only 13 of the nodes carry a material frame; the canopy nodes stay at three degrees of freedom.

6.4 Scaling

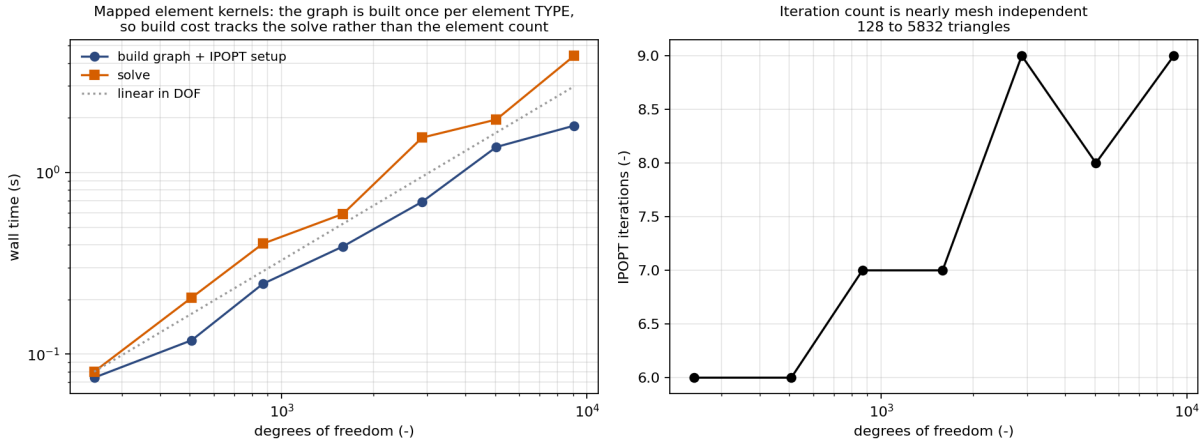


Figure 10: Build and solve time against problem size, and iteration count. Because the graph is built once per element *type*, build cost tracks the solve rather than the element count, and the iteration count is nearly mesh independent across a $37\times$ range in degrees of freedom.

7 Comparison with `kite_fem`

`kite_fem` (with `pyfe3d`) is the existing FEM path in the AWETrim tree. This section states what differs, what was measured, and where the comparison is and is not fair.

7.1 Capability

	<code>kite_fem</code> / <code>pyfe3d</code>	Billow
Elements	springs, pulleys, inflatable Timoshenko beams	cables, pulleys, geometrically exact Timoshenko beams, inflatable tubes, wrinkling membranes
Canopy fabric	noncompressive spring net (no membrane element exists)	CST membrane, Pipkin relaxed energy
Degrees of freedom	6 per node everywhere, masked via <code>bc</code>	3 per node, +3 only on beam nodes
Formulation	Newton–Raphson on $\mathbf{f}_e - \mathbf{f}_i$	minimum potential energy (IPOPT)
Assembly	scipy sparse COO per element	one mapped kernel per element type
Regularisation	<code>I_stiffness=25</code> , <code>step_limit=0.2</code> , <code>relax=0.5</code> , optional pseudo-transient	IPOPT inertia correction; membrane slack blend
Warm start	none (<code>solve</code> resets to undeformed)	state and duals
Default tolerance	10^{-2} N	10^{-8} IPOPT, accepted at 10^{-6} N
Inflatable tube	secant EI/GJ plus collapse flag	the same fits integrated to $W(\kappa)$
Experimental validation	hanging-kite test against measured data	none

7.2 Two traps in making it a fair comparison

1. `FEM.structure.solve` defaults to `I_stiffness=25`, an **absolute** regularisation in N/m added to every degree of freedom of the tangent. The benchmark beams here have $EI/L^3 =$

10 N/m, so the default is larger than the structure: 2000 iterations with the residual stuck at 20 N. With it at zero the same problem converges in 38. It is a kite-scale tuning constant, not a universal one. All results above use zero.

2. The property matching was verified in the linear limit before any conclusion was drawn: below about one degree of rotation the two codes agree to 0.02%.

7.3 Cost and accuracy on a common problem

Tip-loaded slender cantilever, matched properties, same mesh, one machine, one run.

Table 6: Source: `run_cost_comparison.py`. “Warm” is a second solve at a 5% different load; `kite_fem` restarts from the undeformed configuration on every call, so its warm column is a second cold solve.

α	El.	Model	DOF	Setup (s)	Cold (s)	Warm (s)	Iter.	ms/iter
1	20	Billow	126	0.133	0.249	0.081	16	15.6
1	20	<code>kite_fem</code>	126	0.004	0.132	0.151	39	3.4
1	80	Billow	486	0.217	0.733	0.282	17	43
1	80	<code>kite_fem</code>	486	0.009	0.189	0.192	39	4.8
6	40	Billow	246	0.076	0.333	0.062	22	15
6	40	<code>kite_fem</code>	246	0.005	1.384	1.526	444	3.1
6	80	Billow	486	0.186	0.669	0.233	23	29
6	80	<code>kite_fem</code>	486	0.005	2.270	2.658	490	4.8

`kite_fem` has roughly $6\times$ cheaper iterations (compiled `pyfe3d` plus `scipy sparse`) and $30\times$ faster setup. Billow uses 2–20 $\times$ fewer iterations, and its count is nearly independent of both mesh and load (16–23 throughout) where `kite_fem`’s climbs from 39 to about 490 as the load grows. `kite_fem` therefore wins on easy, small, one-shot problems; Billow wins as soon as the problem is hard, reused, or large. Billow’s setup cost is a one-off amortised across a sweep — exactly the coupled-loop pattern.

7.4 The canopy model: spring net versus membrane

Since `kite_fem` has no membrane element, its canopy is a noncompressive spring net (chordwise and spanwise links, *no diagonals*; its own example carries a commented-out block of “fictional springs constraining the canopy from collapse”). Billow can build exactly that net from its cable elements, which makes a genuine like-for-like study possible.

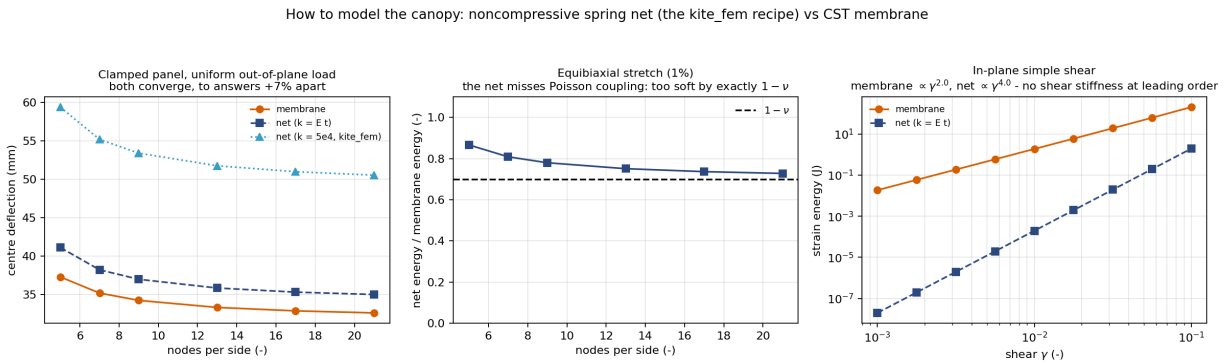


Figure 11: Spring net versus CST membrane on identical meshes, both minimised. Left: both converge under out-of-plane load, to answers 7% apart. Middle: under equibiaxial stretch the net is too soft by exactly $1 - \nu$. Right: under in-plane shear the membrane energy scales as $\gamma^{2.0}$ and the net as $\gamma^{4.0}$ — the net has *no shear stiffness at leading order*.

Measured: biaxial energy ratio 0.728 against $1 - \nu = 0.70$; shear exponents 2.018 (membrane) and 3.999 (net).

7.4.1 Taut: shape and cost

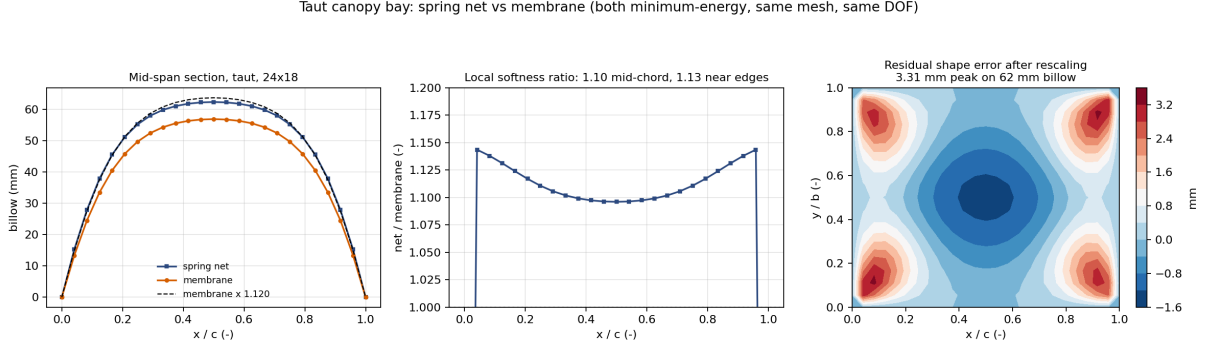


Figure 12: Taut bay, 24×18 , both minimised on the same mesh with the same 1425 degrees of freedom, so this is purely the canopy model. The net billows 9.6% deeper; the best single scale factor is 1.120, and a 3.31 mm residual shape difference remains after removing it.

The local softness ratio is not constant: 1.10 at mid-chord rising to 1.145 near the clamped edges, where the fabric carries biaxial tension and shear that the net cannot represent. The residual map shows the net relatively too deep in the corners and too shallow in the middle — it billows into a slightly different surface, not merely a deeper one.

The scale factor cannot be calibrated away, because it depends on the stress state: about 1.10–1.12 when taut, but 1.00 when fully wrinkled (the slack cases agree within 1%, since a deeply slack panel carries essentially uniaxial chordwise tension where the two models coincide). The correction needed would change with flight condition.

Cost is not a differentiator. At 24×18 : net 0.092 s setup and 0.227 s solve in 12 iterations; membrane 0.176 s and 0.245 s in 8. The membrane’s per-iteration cost is higher but it needs fewer iterations, and the two roughly cancel.

7.4.2 Robustness

Table 7: Peak billow (% chord) and section RMS against each model’s own finest mesh. The net fails outright at one intermediate slack mesh — the signature of its zero-energy shear mode. The membrane never failed.

Mesh	Spring net			Membrane		
	Camber	RMS	Conv.	Camber	RMS	Conv.
<i>taut</i>						
4×3	2.858	0.388	yes	2.542	0.401	yes
8×6	3.129	0.051	yes	2.842	0.051	yes
16×12	3.117	0.010	yes	2.842	0.010	yes
24×18	3.115	—	yes	2.842	—	yes
<i>slack 3%</i>						
8×6	7.388	0.186	yes	7.449	0.143	yes
12×9	7.240	0.091	yes	7.294	0.097	yes
16×12	5.675	1.719	no	7.412	0.031	yes
24×18	7.302	—	yes	7.402	—	yes

7.4.3 Canopy model versus solver, separated

A single bay isolates the two effects that otherwise get conflated. Rows 2 and 3 solve the *identical* spring net, so their difference is the solver alone.

Table 8: One 2.0×1.3 m bay at 250 Pa, all edges fixed.

Case	Configuration	Solve (s)	Iterations	Camber (% c)	Converged
taut 8×6	Billow + membrane	0.030	7	2.842	yes
taut 8×6	Billow + net	0.026	10	3.129	yes
taut 8×6	<code>kite_fem</code> + net	10.04	2729	3.129	yes
taut 12×9	Billow + net	0.034	9	3.083	yes
taut 12×9	<code>kite_fem</code> + net	22.00	3001	5.283	no
slack 8×6	Billow + net	0.049	23	7.388	yes
slack 8×6	<code>kite_fem</code> + net	9.06	3001	7.363	no

Where both converge they agree exactly (3.129% both), which is what makes the `kite_fem` driver used throughout this document trustworthy. On the identical net, minimum energy is roughly $390\times$ faster than the damped Newton solve, and remains convergent where the Newton solve does not.

7.5 Full kite

A V3-scale structure — inflated leading-edge tube, inflated struts at every bay boundary, canopy over nine bays, trailing-edge hem, bridle to a fixed anchor, uniform 250 Pa — was built with the same nodes, tubes and cables in both codes.

Model	DOF	Setup (s)	Cold (s)	Warm (s)	Iter.	Residual (N)
Billow	1893	0.61	5.91	0.58	73	7.2×10^{-9} (converged)
<code>kite_fem</code> ($I=25$)	2976	0.05	76.2	72.0	2001 (cap)	4.1×10^5 (diverged)
<code>kite_fem</code> ($I=0$)	2976	0.06	108.9	68.9	2001 (cap)	1.3×10^8 (diverged)

Interpretation, carefully. No usable `kite_fem` full-kite number was obtained. Diagonals, canopy curvature, both regularisation settings and load stepping with rebuilt initial conditions were all tried; all diverged, with residuals roughly $100\times$ the applied load. **This is consistent with a known open issue on the `kite_fem` side** rather than a defect in the driver used here — the ASKITE integration commit records “`kite_fem` still diverging, but not due to the coupling”. The only kite-scale figure that comes from its own workflow is its hanging-kite validation, at 442–671 s per load case. The single-bay comparison above is the trustworthy cost comparison.

8 How many membrane triangles?

Chordwise and spanwise resolution are swept independently, because they are not the same problem: a canopy bay billows across the chord — which sets the section the aerodynamics sees — while spanwise it varies slowly between struts.

The test case is one bay of the LEI-V3 canopy: chord 2.0 m, strut spacing 1.3 m (the V3 spans 8.28 m over 10 stations), held on all four edges, under 250 Pa. Both a taut panel and a realistically slack one (reference cut 3% oversize) are run.

Convergence is judged on quantities the coupling actually consumes: peak billow over chord, and the RMS deviation of the mid-span chordwise profile from the finest mesh. **Peak local strain is deliberately not a criterion:** in a wrinkling membrane it does not converge, and nothing downstream reads it.

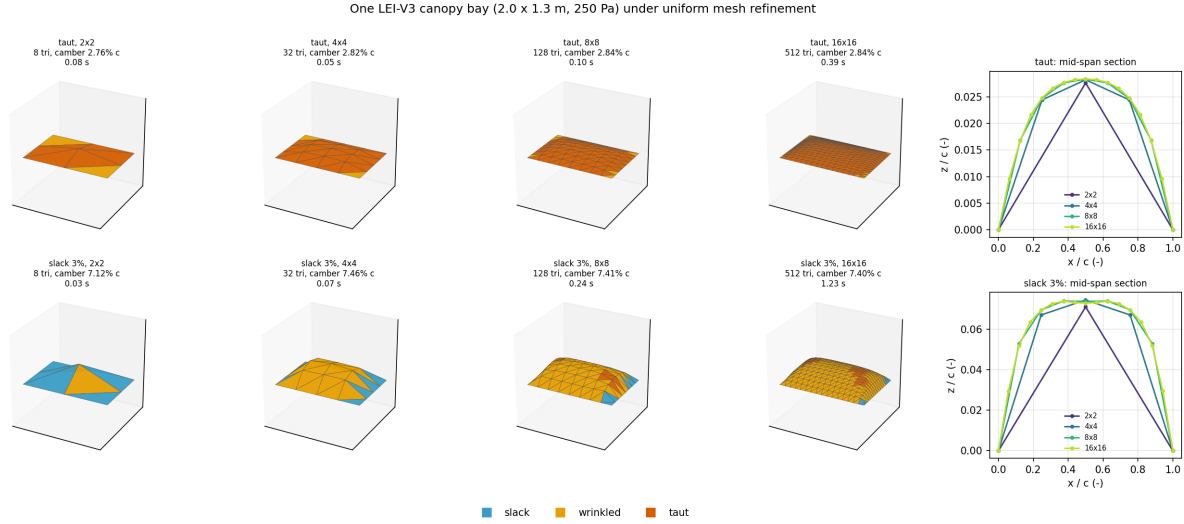


Figure 13: One bay under uniform refinement. A 2×2 bay is a tent, not a canopy: the camber is only 2.8% low but the section is two straight lines. By 8×8 the profile is visually converged.

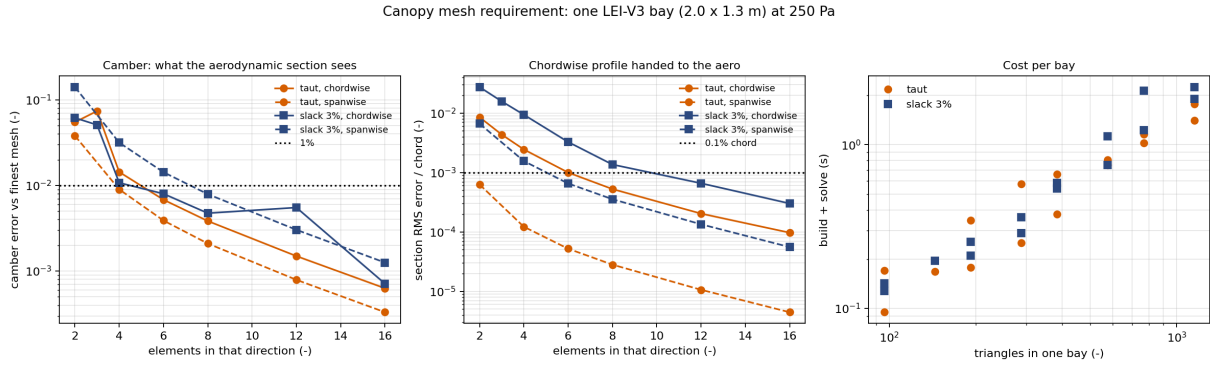


Figure 14: Camber and section-shape convergence, chordwise (solid) and spanwise (dashed), and cost per bay.

Table 9: Candidate meshes against a 24×24 reference, slack 3%.

Mesh	Triangles	Camber error	Section RMS (% c)	Build + solve (s)	Iter.
4×4	32	0.88%	0.967	0.03	12
6×6	72	0.61%	0.338	0.07	17
8×6	96	0.71%	0.145	0.09	19
8×8	128	0.16%	0.140	0.17	15
12×8	192	0.18%	0.075	0.18	21
16×12	384	0.21%	0.033	0.44	22

Recommendation. **8 chordwise $\times$ 6 spanwise per bay**, 96 triangles, about 900 for the whole nine-bay canopy. Camber error 0.71% and section RMS 0.145% of chord — 2.9 mm on a 2 m chord, comfortably below what an airfoil parameterisation resolves. Use 12×8 for final runs: it halves the shape error for twice the cost.

Slack costs roughly twice the mesh. At 3% oversize the panel is 88% wrinkled (against 1–4% taut) and needs about 12×8 for the same accuracy, with 4 $\times$ the iterations. Since a real

canopy flies slack, the as-cut slack of the actual sail drives the mesh requirement more than anything else here and is worth pinning down.

Caveat. These numbers are for *uniform* pressure. A real chordwise pressure distribution is peaked near the leading edge and may demand more chordwise resolution; the study should be repeated once loads come from an aerodynamic solve.

9 Known limitations

No experimental validation. Every check in this document is against closed-form solutions, published benchmarks, or the ASKITE fits. The claim that the membrane is “more correct” than a spring net rests on it being the physically appropriate model, not on measurement. `kite_fem` has a hanging-kite test against real data; Billow has nothing equivalent yet.

Applied moments. Need load stepping, and are exact only about a fixed axis; a dead three-dimensional moment is non-conservative and has no potential. Nodal forces are unaffected.

Post-collapse tubes. Reported, not modelled. The fits cover bending and torsion only; axial and shear stiffness must be supplied from tube geometry.

Dead loads. Loads are frozen per solve, which matches a staggered coupling loop. A true follower pressure would not be conservative and could not be posed as a potential — worth remembering before anyone tries to fold aerodynamics into the same minimisation.

No contact. Contact and self-contact are not modelled.

Setup cost. About 0.2s of graph construction for a 486-DOF beam is disproportionate against `kite_fem`’s 0.005s, even though it is a one-off and stops mattering at membrane scale.

Threaded assembly. `PotentialEnergy(parallelization=...)` exposes CasADi threaded maps but is untested at scale; serial is the default.

Not coupled. There is no aerodynamic coupling adapter yet.

10 Reproducing everything in this document

Figure / table	Command
Figs. 7–10, 11	<code>python scripts/structural/run_demo_cases.py</code>
Figs. 1–4	<code>python scripts/structural/run_validation_benchmarks.py</code>
Cantilever cost table	<code>python scripts/structural/run_cost_comparison.py</code>
Figs. 13, 14	<code>python scripts/structural/run_mesh_requirement.py</code>
Single-bay comparison	<code>python scripts/structural/run_panel_cost.py</code>
Full-kite table	<code>python scripts/structural/run_full_kite_cost.py</code>
Test suite	<code>pytest tests/structural/</code>

10.1 Package layout

<code>rotations.py</code>	<i>SO</i> (3) kernels over an <code>xp</code> namespace (NumPy or CasADi)
<code>model.py</code>	<code>DofLayout</code> , <code>StructuralState</code> , <code>StructuralModel</code>
<code>energy.py</code>	<code>PotentialEnergy</code> : mapped assembly, gradient, tangent stiffness
<code>solver.py</code>	<code>MinimumEnergySolver</code> , <code>StructuralSolution</code>
<code>elements/base.py</code>	<code>ElementKernel</code> protocol, <code>ElementSet</code>
<code>elements/cable.py</code>	<code>CableKernel</code> , <code>PulleyKernel</code>
<code>elements/beam.py</code>	<code>TimoshenkoBeamKernel</code> , <code>BeamSection</code>
<code>elements/inflatable.py</code>	<code>InflatableTubeLaw</code> , <code>InflatableBeamKernel</code>
<code>elements/membrane.py</code>	<code>MembraneKernel</code> , <code>membrane_regimes</code>

10.2 Minimal example

```
import numpy as np
from awetrim.structural import MinimumEnergySolver, StructuralModel
from awetrim.structural.elements import build_cable_elements

nodes = np.array([[0., 0, 0], [1., 0, 0]])
cables = build_cable_elements([[0, 1]], rest_lengths=[1.0], stiffness=[1e4])
model = StructuralModel(nodes, [cables], fixed_translation_nodes=[0])
solution = MinimumEnergySolver(model).solve(np.array([[0., 0, 0], [10., 0, 0]]))
```

10.3 Invariants to preserve

- **Isolation.** NumPy and CasADi only.
- **One mapped kernel per element type.** Never reintroduce a Python loop over elements that builds MX.
- **Single source for every formula.** `rotations.py` holds the $SO(3)$ maps against an `xp` namespace so NumPy and CasADi share one implementation; `beam_strains` and `green_strain` are called by both the kernels and the NumPy setup and diagnostic paths.
- **No CasADi across the module boundary.** Symbolics are created inside `energy.py` and stay there.
- **Kernels are pure.**
- **converged is the physical verdict**, with IPOPT's own kept alongside.

References

- [1] J.C. Simo and L. Vu-Quoc, “A three-dimensional finite-strain rod model. Part II: computational aspects”, *Comput. Methods Appl. Mech. Engrg.* **58** (1986) 79–116.
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